

Assessing the Water Footprint of Large Language Models

Chosen topic: Assess the Environmental Impact of Large Language Models Beyond Energy Consumption

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Introduction

In 2023, data centers specifically dedicated to Large Language Models (LLM) and other AI applications consumed an estimated 30 billion liters of water for cooling purposes. Driven by the exponential growth of AI applications, projections suggest this figure could escalate to 338 billion liters by 2030 (Gröger et al., 2025).

The industry is already facing severe challenges regarding water-based cooling infrastructure and the physical consequences of this high water demand are becoming visible in local communities. For instance, residents in Georgia (USA) have reported significant drops in water pressure since a data center opened nearby (Skibell, 2026).

These incidents highlight that water supply is no longer just a logistical detail, but a crucial ecological bottleneck. Additionally, there is also the question of whether it always has to be water, and whether water cooling methods could be made more effective by a better choice of data center location or by using more resource-efficient LLM models.

Water Cooling in Data Centers

To understand this challenge, we must distinguish between water usage and consumption. Data center water consumption falls into three scopes: direct on-site usage, external usage for electricity generation, and indirect usage for hardware production (Li et al., 2025). This work focuses specifically on Scope 1: direct water consumption inside facilities.

While water usage implies the resource returns to its local source, true consumption permanently removes it from the environment. In data centers, this occurs primarily through evaporation in cooling towers and chillers (Hiremath, 2024). Consequently, GPT-3 consumes around 0.5 liters of water per 10–50 requests, depending on data center location (Li et al., 2025). Furthermore, cooling water can only be recycled 3–4 times before mineral accumulation requires its disposal.

Water-free methods like air cooling are generally only effective for small data centers in cold climates (Hiremath, 2024). Additionally, they often shift the water footprint from direct on-site

use (Scope 1) to indirect electricity generation (Scope 2), displacing rather than mitigating the problem.

In response, advanced water-free technologies are emerging. Immersion cooling submerges servers in dielectric fluid (Simon, 2018), while direct-to-chip cooling circulates coolant in a closed loop, saving over 125 million liters annually (Ahmadi et al., 2026). Both prove water-free cooling is feasible for AI hardware, though their environmental impacts require further analysis.

Water footprint on LLMs

A common metric for the direct water consumption of data centers is the Water Usage Effectiveness (WUE), defined as the onsite water volume evaporated per kilowatt hour of IT equipment energy:

$$\text{WUE [L/kWh]} = \frac{\text{Annual Site Water Usage}}{\text{ICT Equipment Energy}}$$

While WUE is often criticized for ignoring off site (Scope 2 and 3) water use (Mytton, 2021), it remains a useful indicator for direct cooling related water demand. This is particularly relevant for LLMs deployed in GPU accelerated clusters.

Recent benchmarking of 30 LLMs shows that the water footprint per query varies dramatically with model choice and infrastructure. The most water intensive model consumes over 200 mL of fresh water per long query. Scaling a single model like GPT 4o to 700 million daily queries would evaporate an amount equivalent to the annual drinking water needs of 1.2 million people. Encouragingly, the same model can reduce its water footprint by 85 percent simply by being deployed on more efficient hardware (Singh et al., 2025).

These per query numbers are consistent with bottom-up energy and water models for United States data centers. In 2023, the average site WUE of United States data centers was about 0.36 L/kWh, and the total direct water consumption reached 66 billion liters, with hyperscale and colocation facilities accounting for 84 percent of that amount. By 2028, the average WUE is projected to rise to 0.45 to 0.48 L/kWh, driven by the increasing use of liquid cooled AI servers. These servers are energy efficient (low PUE) but water intensive (Hamm, 2025). However, this projection assumes no rapid adoption of closed-loop water-free cooling (e.g., immersion or direct-to-chip). If those technologies mature faster, the WUE could plateau or even decline.

Moreover, the indirect water footprint from electricity generation is about an order of magnitude larger than the direct one. United States data centers consumed roughly 800 billion liters of indirect water in 2023, corresponding to 4.52 L/kWh, slightly above the national average of 4.35 L/kWh (Hamm, 2025). For LLMs, which require massive and often latency sensitive compute, the total water burden therefore depends not only on the model architecture and query complexity (Sundaram et al., 2025) but also on the regional electricity mix and the cooling strategy chosen. Balancing energy efficiency (PUE) and water efficiency (WUE) remains a key challenge for sustainable LLM deployment. A recent approach by Moore et al. (2025) addresses this via SLIT, a scheduler for geo-distributed LLM inference that co-optimizes water, carbon, cost, and response time. Compared to state-of-the-art systems (Helix, Splitwise), SLIT reduces water use by up to 99%, carbon by 98%, and energy cost by 99% while maintaining competitive latency. This demonstrates that intelligent request routing can greatly cut direct water consumption

(Moore et al., 2025). But it has to be said that under idealized conditions (real network latencies and load) distribution could reduce the savings.

Conclusion

In conclusion, the rapidly growing water footprint of LLMs presents a critical ecological challenge. While direct on-site water cooling strains local resources, simple air-cooled alternatives often merely displace the burden to indirect energy generation. Achieving sustainable AI infrastructure requires a holistic approach. This demands the adoption of advanced, water-free cooling technologies like immersion or direct-to-chip cooling, strategic geographical placement of data centers, and the continuous optimization of both model architectures and hardware deployment efficiency.

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